

Beyond Task Activation: A Vigour-Related Low-Variability Network in Parkinson's Disease

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2 ABSTRACT

3 Movement vigour is reduced and behaviourally unstable in Parkinson's disease (PD), but it
 4 remains unclear whether trial-by-trial fluctuations in vigour are represented by a stable network-
 5 level signal that can be distinguished from broader task engagement. We analysed task-fMRI
 6 data from patients with PD performing a reward-cued ballistic hand-squeeze task and used
 7 reaction time as a trial-wise index of vigour. Trial-wise GLM beta estimates were entered into a
 8 constrained optimization framework that favoured task engagement, behavioural relevance, spatial
 9 smoothness, and low consecutive-trial variability. Across ablation settings, the full model yielded
 10 the lowest mean evaluation score ($S = 0.0164$) and outperformed 8 of the 11 alternative settings
 11 after Bonferroni correction. The resulting projection showed lower consecutive-trial variability than
 12 reaction time across 56 runs from 14 subjects ($\beta = -43.94$, $z = -16.52$, $p = 2.55 \times 10^{-61}$). The final
 13 map comprised 2,674 voxels, with a motor-related core in peri-central cortex, superior parietal
 14 cortex, the left putamen, and right cerebellar regions, together with strong ROI-normalized
 15 representation in temporal-limbic regions. Spatial overlap with a matched task-engagement
 16 reference map was limited (102 shared voxels, Dice coefficient = 0.049, center-of-mass distance
 17 = 47.3 mm). Network topology was more similar across subjects in the ON than OFF medication
 18 state, whereas galvanic vestibular stimulation effects were regionally selective and heterogeneous
 19 across participants. These findings identify a distributed candidate vigour-related network in
 20 PD whose trial-wise expression is relatively stable despite behavioural variability and whose
 21 organization is sensitive to dopaminergic state.

22 **Keywords:** Parkinson's disease, fMRI, motor vigour, variability, functional connectivity

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1 INTRODUCTION

24 Movement vigour refers to how quickly and energetically an action is initiated and carried out. It is not
 25 fixed: the same person can perform the same action faster or slower across trials, and different people often
 26 express different levels of speed and force during the same task (Berret et al., 2018). Work in humans and
 27 animals shows that vigour is shaped by a distributed motor network that includes motor and premotor
 28 cortex, supplementary motor area, basal ganglia, and thalamus (Anderson et al., 2012; Spraker et al., 2007a;
 29 Shirinbayan et al., 2019a). This suggests that vigour is not merely a behavioural outcome, but a regulated
 30 feature of movement controlled by interacting neural circuits.

31 Within this network, the basal ganglia play an important role in setting movement speed and amplitude, as
 32 well as in selecting which action to perform. Reducing basal ganglia output can make movements smaller
 33 and slower without changing the chosen action (Turner and Desmurget, 2010), and manipulating striatal
 34 pathways can speed up or slow down movement in a dopamine-dependent way (Yttri and Dudman, 2016).

Reward and dopamine also influence vigour: actions linked to higher value are often initiated faster and executed with greater speed (Kawagoe et al., 1998; Reppert et al., 2015; Summerside et al., 2018). Together, these findings show that vigour is shaped by cortico-basal ganglia circuits and motivational context, and that dopamine is central to this regulation.

This role of dopamine makes Parkinson's disease (PD), which is characterised by dopaminergic loss in the basal ganglia, a useful model for studying vigour. A core motor symptom of PD is bradykinesia, typically described as slower and smaller movements. The problem, however, is not simply that patients move slowly. When speed is strongly encouraged, patients with PD can sometimes move quickly (Mazzoni et al., 2007a), indicating that the motor system is not fundamentally incapable of producing fast actions. Rather, PD may disrupt how vigour is regulated from one trial to the next. Several lines of evidence support this idea. Behavioural studies report larger short-term fluctuations in response time in PD (MacDonald et al., 2025), and earlier work found increased endpoint variability that can be partly reduced when patients slow down or rely on visual guidance (Sheridan and Flowers, 1990).

Motor output itself is also less stable in PD: motor unit discharge becomes more variable in mild-to-moderate disease, even when mean discharge rates remain relatively preserved (Wilson et al., 2020). Beyond these output measures, dopamine loss appears to affect both average movement features and movement irregularity, with different striatal regions linked to speed, amplitude, and kinematic consistency (Lee et al., 2015). Together, these findings suggest that dopamine shapes not only how much vigour is expressed, but also how consistently it is expressed across repeated actions.

The unresolved issue is where this variability enters the motor system. Variability in movement can arise from several sources. Some of it is useful, helping the motor system explore actions, learn from reward, and adjust after errors (Pekny et al., 2015; Wu et al., 2014). Other variability reflects the neural state before movement; for example, preparatory activity in motor and premotor cortex can predict the speed of the upcoming reach (Churchland et al., 2006). A further important source is motor noise (Faisal et al., 2008), which is not always beneficial. This noise is signal-dependent, meaning that larger motor commands generate greater motor noise and variability, so faster or more forceful movements become harder to control accurately (Harris and Wolpert, 1998). High motor noise can also impair learning by weakening the link between an action and its outcome, as shown in work on cerebellar damage (Therrien et al., 2016).

Because these sources occur together, behaviour alone cannot reveal whether variable vigour reflects an unstable central signal or noise added later in the motor pathway. This distinction is important for PD: if the central vigour signal itself is unstable, then bradykinesia and inconsistency may reflect impaired neural encoding; if a relatively stable brain signal still tracks behaviour while behavioural output fluctuates, then part of the instability may arise downstream of that signal. Recent fMRI work speaks to this question (Dinstein et al., 2015; Garrett et al., 2013; Haar et al., 2017), but it has focused mainly on average vigour, broad motor variability, or activity in single regions. A gap therefore remains in understanding whether trial-wise fluctuations in vigour are represented by a stable, low-variability network-level pattern.

To test this distinction, we analysed task-fMRI data from patients with Parkinson's disease performing a ballistic hand-squeeze task. We used reaction time (RT) as the main trial-wise index of movement vigour because it provided the most reliable behavioural measure across participants and runs. We then applied a constrained optimization framework to identify a distributed brain pattern whose activity tracked reaction time across trials while showing low trial-to-trial variability. We hypothesised that, despite substantial variability in behaviour and in single-region responses, vigour fluctuations would be captured by a distributed and relatively stable network that overlaps with known vigour-related circuitry. We further expected this network signature to explain behavioural variability beyond mean task-evoked activity, helping to separate central vigour-related coding from variability that may be added later in the motor pathway.

2 MATERIALS AND METHODS

2.1 Participants

Fifteen participants with Parkinson's disease were included in the final analysis (4 female, 11 male; mean age 68.9 ± 6.35 years, range 58-82). Eleven participants were right-handed, and four were left-handed.

One additional participant was excluded from analysis because one recording run was excessively noisy. Exclusion criteria included prior neurosurgical treatment for Parkinson's disease, other neurological or major medical conditions, dementia severe enough to preclude informed consent, severe levodopa-induced dyskinesia incompatible with MRI, and standard contraindications to MRI. Clinical severity was assessed using the MDS-UPDRS (Goetz et al., 2008), and medication status was recorded in both OFF and ON dopaminergic medication states. All participants provided written informed consent, and the study was approved by the Clinical Research Ethics Board at the University of British Columbia.

2.2 Behavioural task

During fMRI, participants performed a ballistic hand-squeeze task implemented in MATLAB with Psychtoolbox (Brainard and Vision, 1997) while lying supine in the scanner and viewing the task through a mirror display (Figure 1). On each trial, a visual cue indicating reward value was presented for 2 s, followed by a variable delay of 2–3.5 s. A moving balloon then appeared as the Go signal. Participants were instructed to squeeze a mechanical bulb as quickly and forcefully as possible when the Go signal appeared, and successful responses were followed by a visual burst of the balloon.

Participants completed two medication sessions on the same day: OFF and ON medication. Participants were instructed to stop their dopaminergic medication at least 12 h before recording. The order was fixed across participants: the OFF medication session was always completed first, after which participants took their medication and completed the ON medication session. Each medication session consisted of two task runs, and each run included nine blocks of 10 trials (90 trials per run). Most trials were rewarded, with low- and high-reward conditions counterbalanced across runs, and 10% were catch trials in which participants were required to withhold their response to avoid a penalty. Reward was included to maintain task engagement; however, analyses in the present study focused on motor vigour irrespective of reward condition. Task difficulty was adjusted online by varying the response window on the basis of recent performance, indexed by hit rate, to maintain an accuracy of at least 65%. A 20 s rest period was provided after every three blocks.

Reaction time (RT), defined as the interval between Go-signal onset and squeeze initiation, was considered the primary trial-wise index of motor vigour because it was more robust across participants and runs than other candidate behavioural measures, such as time to peak force and squeeze duration.

2.3 Galvanic vestibular stimulation

Galvanic vestibular stimulation (GVS) was delivered using an STMISOLA constant-current stimulator (BIOPAC Systems Inc.) through 41-mm Ag/AgCl electrodes placed bilaterally over the mastoid processes. Electrode placement was verified using a 1 Hz sinusoidal stimulus, and stimulation amplitude was individualized using a staircase procedure to determine the highest current below the threshold for superficial cutaneous sensation at the electrode site (Nguyen et al., 2022).

Eight active amplitude-modulated waveforms were generated in MATLAB using suppressed-carrier modulation. The waveforms varied in carrier frequency (75 or 125 Hz) and envelope frequency (0, 10, 16, or 20 Hz). The resulting GVS stimuli were labeled as follows: GVS1, 75 Hz carrier and 0 Hz envelope; GVS2, 75 Hz carrier and 10 Hz envelope; GVS3, 75 Hz carrier and 16 Hz envelope; GVS4, 75 Hz carrier and 20 Hz envelope; GVS5, 125 Hz carrier and 0 Hz envelope; GVS6, 125 Hz carrier and 10 Hz envelope; GVS7, 125 Hz carrier and 16 Hz envelope; and GVS8, 125 Hz carrier and 20 Hz envelope. A sham condition, in which no current was delivered (0 mA), was also included.

These frequencies were selected to approximate typical vestibular afferent firing rates and oscillatory bands implicated in motor planning and execution.

2.4 MRI acquisition and preprocessing

Structural and functional MRI data were acquired on a Philips 3T Ingenia Elition X whole-body MRI scanner upgraded to an MR7700 system (Philips Healthcare, Best, The Netherlands) using a 32-channel head coil. For each participant, a T1-weighted structural image was acquired, followed by two task-based fMRI scans, one for each run of the behavioural task.

T1-weighted images were acquired using a 3D MPRAGE sequence with the following parameters: flip angle = 8°, echo time (TE) = 3.6 ms, repetition time/shot interval (TR) = 2400 ms, inversion time = 950 ms,

180 slices, field of view (FOV) = $256 \times 256 \times 180 \text{ mm}^3$, and spatial resolution = 1 mm^3 isotropic. Images were acquired with a compressed SENSE factor of 3.6, and the acquisition time was 3:42 min.

Functional images were acquired using echo-planar imaging (EPI). For all participants, single-echo images were acquired using the following parameters: flip angle = 77° , TE = 30 ms, TR = 1000 ms, 51 slices, FOV = $240 \times 153 \times 240 \text{ mm}^3$, spatial resolution = 3 mm^3 isotropic, parallel imaging with SENSE factor = 2 and multiband factor = 3, and acquisition time = 14:17 min. Two spin-echo EPI sequences were also acquired with opposite fat-shift directions, anterior–posterior and posterior–anterior, for distortion correction. These images were acquired with the following parameters: flip angle = 77° , TE = 55 ms, TR = 2650 ms, 51 slices, FOV = $240 \times 153 \times 240 \text{ mm}^3$, spatial resolution = 3 mm^3 isotropic, parallel imaging with SENSE factor = 2 and multiband factor = 3, and acquisition time = 5.3 s.

Image preprocessing was performed in FSL (Smith et al., 2004). Structural images underwent bias-field correction and brain extraction. Functional images were corrected for head motion and susceptibility-induced distortions, with distortion correction referring to correction of susceptibility-related spatial distortions in the functional images using the spin-echo images acquired with opposite fat-shift directions. No initial functional volumes were discarded during preprocessing because the scanner automatically discarded four dummy volumes before the start of saved image acquisition. Functional images were then high-pass filtered using a 100 s cutoff, brain extracted, denoised using ICA-AROMA (Pruim et al., 2015), and spatially smoothed with a 5 mm full-width at half-maximum Gaussian kernel. The preprocessed functional images were then registered to each participant's structural image and normalized to standard MNI152 space.

2.5 Trial-wise GLM estimation and beta preprocessing

Task-related responses for each modeled trial were estimated using trial-wise general linear model (GLM) beta estimates with GLMsingle (Kay et al., 2013; Prince et al., 2022). Analyses were performed separately for each subject-session. Within each subject-session, the two task runs were analyzed jointly while preserving run identity in the design matrix and nuisance regressors. The GLM design matrix included task-trial onset regressors with a fixed trial duration of 9 s. Catch trials were not modeled as separate regressors, and reward-condition labels were not included in the GLM design. To restrict the analysis to relevant tissue, voxels corresponding to background, cerebrospinal fluid (CSF), and white matter were masked out, retaining only gray-matter voxels. The initial MNI gray-matter, non-CSF candidate mask contained 91,781 voxels. The mean CSF time series for each run was included as a nuisance regressor.

GLMsingle was run with HRF fitting, GLMdenoise, and fractional ridge regression enabled. HRF fitting used the default GLMsingle library of 20 candidate HRFs, and the HRF yielding the highest variance explained was selected separately for each voxel. GLMdenoise derived additional nuisance regressors from low task-related-variance voxels using principal components analysis and selected the number of noise components by cross-validation. Fractional ridge regression was used to stabilize single-trial beta estimates; shrinkage fractions from 0.05 to 1.00 in increments of 0.05 were evaluated, and the optimal shrinkage level was selected separately for each voxel by cross-validation. Run-specific baseline terms were modeled using GLMsingle's run-wise polynomial regressors, including degree-0 intercepts and higher-order polynomial terms up to degree $\text{round}(L/2)$ for a run of length L minutes.

Before applying GLMsingle, high-DVARS trials were identified from the preprocessed BOLD time series. For each file, the DVARS metric was robustly standardized using a median absolute deviation (MAD)-based score (z_i). Trials with $z_i > 3$ were labeled as high-outlier trials. The number of trials removed by this criterion was capped at 15% per file to avoid excessive loss of trials while removing the largest signal-change outliers.

A per-session preprocessing step was then applied to the GLMsingle beta estimates. To detect outlier beta values, they were robustly normalized using the session-level median and MAD. Beta values with $|\beta_{\text{norm}}|$ above the 99.9th percentile were flagged as beta outliers. For each voxel, the beta-outlier fraction was computed as the proportion of trials flagged as outliers. Voxels were retained when this fraction was ≤ 0.5 and excluded when more than 50% of their trials were beta outliers.

The run-level preprocessing outputs retained $84,004 \pm 2,696$ voxels on average. This corresponded to an average removal of $8.47\% \pm 2.94\%$. To assess whether this filtering disproportionately affected susceptibility-prone ventral regions, removal rates were summarized using Harvard-Oxford atlas labels and

ventral-position proxies. Average run-level removal was 8.83% in frontal orbital cortex, 8.86% in temporal pole, 9.75% in inferior temporal gyrus, 8.80% in the ventral bottom quartile, and 8.44% in all other voxels. Thus, run-level preprocessing did not disproportionately remove orbitofrontal, temporal pole, or ventral voxels.

The cleaned outputs from all subjects, sessions, and runs were then used to define a shared voxel space across the dataset. The coverage threshold was applied at the trial level across all concatenated trials from all subjects, sessions, and runs, rather than separately within each subject or session. Voxels were retained in the final shared mask if they had finite beta estimates for at least 70% of the dataset-wide trial slots. The final shared group mask contained 91,601 voxels, yielding a final group beta matrix of 91,601 voxels $\times$ 4,980 trials. Therefore, the group-level 70% coverage filter removed only 180 voxels from the initial 91,781-voxel candidate mask, retaining 99.80% of candidate voxels and excluding 0.20%.

The final shared-mask pruning was slightly enriched in ventral/temporal susceptibility-prone regions, but the absolute effect was small: 61 voxels were removed from a susceptibility-prone proxy set of 10,056 voxels. Orbitofrontal cortex itself was not disproportionately removed, with 2 of 1,474 orbitofrontal voxels excluded, corresponding to 0.14%. Because voxels with many outlier or missing beta estimates may reflect less stable estimates, this procedure could, in principle, bias the retained mask toward more stable voxels. However, the final group-level coverage filter had a minimal effect on the shared voxel mask, removing only 180 voxels. The larger filtering effect occurred during run-level preprocessing, and the voxel-retention statistics above were therefore reported as quality-control metrics. All subsequent analyses were performed on the concatenated cleaned beta data in the final shared voxel space.

2.6 Optimization framework

A group-level optimization procedure was used to identify a vigour-related brain network. The goal of the model was to find a linear weighted voxel-activation pattern (w), whose projection onto the neural data produces a one-dimensional neural signal, y . This projected signal was defined as

$$y = w^T X, \quad (1)$$

where X denotes the neural data matrix and w contains the voxel weights. Depending on the specific loss term, the projection was applied either to the beta estimates or to the BOLD signal:

$$y_\beta = w^T X_\beta, \quad y_{\text{bold}} = w^T X_{\text{bold}}. \quad (2)$$

Here, X_β denotes the beta-estimate matrix and X_{bold} denotes the BOLD signal matrix.

Because of the high dimensionality of the data and the presence of many missing values after the preprocessing steps described above, the BOLD signals were first reduced using expectation-maximization principal component analysis (EMPCA) (Roweis, 1997). The number of components was determined using an 80% cumulative explained-variance threshold, which corresponded to 100 components in this study. The resulting PCA weights were then applied to the beta estimates, projecting them into the same PCA space. The optimization was therefore performed in the reduced PCA space, and the final weights were later projected back into voxel space.

The optimization model was based on four assumptions about the neural representation of motor vigour. First, the target network should consist of voxels engaged during task performance. Second, the projected signal should encode behavioural information. Third, neighbouring voxels should show similar contributions, reflecting spatial coherence. Fourth, the selected signal should show low consecutive-trial variability. These four conditions were incorporated into the model one by one, as described below.

Condition 1: The projected signal should come from task-engaged voxels.

The first condition was that the projected signal should be generated mainly from voxels that were engaged during task performance. Each voxel contributes to y_β through its beta activity and its corresponding weight in w . Therefore, if a voxel has weak task-related beta activation, assigning a small weight to that voxel

230 would allow a high task-engaged region to contribute strongly to the projected signal:

$$\min_w \sum_j \frac{w_j^2}{|X_{\beta,j}|} = \min_w w^\top \text{diag}\left(\frac{1}{|X_\beta|}\right) w.$$

231 This gives the task-engagement loss term:

$$\mathcal{L}_{\text{task}}(w) = w^\top C_{\text{task}} w, \quad C_{\text{task}} = \text{diag}\left(\frac{1}{|X_\beta|}\right).$$

232 Thus, C_{task} favours voxels with stronger task-related beta responses and discourages the model from
233 selecting voxels with weak task engagement.

234 **Condition 2: The projected signal should encode behavioural information.**

235 The second condition was that the projected neural signal should be behaviourally relevant. In this study,
236 the target behavioural variable was reaction time. Therefore, the model should find w such that y is strongly
237 associated with the behavioural variable X_{behav} :

$$\begin{aligned} \max_w \text{Corr}(X_{\text{behav}}, y_\beta)^2 &= \max_w \text{Corr}(X_{\text{behav}}, w^\top X_\beta)^2 \\ &= \max_w \frac{[(w^\top X_\beta) X_{\text{behav}}]^\top}{w^\top (X_\beta X_\beta^\top) w} \\ &= \max_w \frac{w^\top (X_\beta X_{\text{behav}}^\top X_{\text{behav}} X_\beta^\top) w}{w^\top (X_\beta X_\beta^\top) w} \\ &= \max_w \frac{w^\top C_b w}{w^\top C_d w}, \end{aligned}$$

238 where

$$C_b = X_\beta X_{\text{behav}}^\top X_{\text{behav}} X_\beta^\top, \quad C_d = X_\beta X_\beta^\top.$$

239 Here, C_b captures the component of the projected beta signal that is aligned with behaviour, while C_d
240 normalizes the association by the variance of the projected beta signal. Since the final optimization problem
241 is written as a minimization problem, this behavioural-relevance term is later included with a negative sign.

242 **Condition 3: The projected signal should come from a spatially coherent network.**

243 The third condition was that the selected pattern should form a spatially coherent network rather than a
244 scattered set of isolated voxels. It means neighbouring voxels were encouraged to have similar weights. If
245 $\mathcal{E}$ denotes the set of neighbouring voxel pairs, this penalty can be written as

$$\min_w \sum_{(p,q) \in \mathcal{E}} (w_p - w_q)^2. \quad (3)$$

246 A spatial adjacency matrix was constructed from voxel anatomy, and its graph Laplacian was used to define
247 the spatial smoothness penalty. This gives

$$\mathcal{L}_{\text{smooth}}(w) = w^\top C_{\text{smooth}} w, \quad C_{\text{smooth}} = L_{\text{graph}}. \quad (4)$$

248 Thus, C_{smooth} enforces spatial smoothness across neighbouring voxels. When the optimization is performed
249 in PCA space, this spatial smoothness penalty is expressed in the same reduced space before optimization.

Condition 4: The projected signal should show low consecutive-trial variability.

The fourth condition was that the selected signal should exhibit low consecutive-trial variability. Variability was therefore defined as the difference between consecutive trials, rather than as variance across the full time series. This distinction is important because variance captures the overall dispersion of a signal around its mean but does not account for trial order. As a result, two projected signals may have similar variance even if one evolves smoothly across trials while the other fluctuates sharply from one trial to the next. Consecutive-trial differences, in contrast, preserve the local temporal structure of the signal and directly measure abrupt changes between neighbouring trials.

This definition is motivated by the behavioural structure of motor vigour. Vigour on trial i is unlikely to be statistically independent from vigour on trial $i + 1$, because consecutive trials are shaped by similar physiological, attentional, and task-related states. Trials that occur close in time are therefore expected to be more similar than trials separated by longer intervals. Accordingly, a stable vigour-related neural signal should not contain large unexplained jumps between consecutive trials. This reasoning is consistent with previous work on reaction-time variability showing that standard deviation alone does not capture temporal dependency, whereas measures such as lag-one autocorrelation and successive-trial differences reveal the local structure of performance fluctuations (Perquin et al., 2023).

A similar principle has been applied in fMRI research using mean squared successive difference (MSSD) and related measures, which quantify moment-to-moment changes in BOLD activity rather than dispersion around a global mean (Garrett et al., 2011; Wehrheim et al., 2024; Nomi et al., 2018). This approach is particularly appropriate for task fMRI, where the expected signal may change gradually over time, across task conditions, or between blocks. A global variance measure could incorrectly treat such slow task-related changes as variability, whereas a consecutive-difference penalty specifically targets local instability between consecutive trials. Thus, the low-variability term in the present model penalized large consecutive-trial changes in the projected signal while allowing slower task-related changes to remain. Boundaries between runs and blocks were excluded from this penalty because trials across these boundaries do not reflect continuous local temporal dynamics and may contain artificial discontinuities introduced by changes in run structure or task context.

We need to minimize variability between consecutive trials:

$$\begin{aligned} \min_w \sum_{i \in \mathcal{I}_c} \|y_{\text{bold},i} - y_{\text{bold},i+1}\|_2^2 &= \min_w \sum_{i \in \mathcal{I}_c} (y_{\text{bold},i} - y_{\text{bold},i+1}) (y_{\text{bold},i} - y_{\text{bold},i+1})^\top \\ &= \min_w \sum_{i \in \mathcal{I}_c} \left(w^\top X_{\text{bold},i} - w^\top X_{\text{bold},i+1} \right) \left(w^\top X_{\text{bold},i} - w^\top X_{\text{bold},i+1} \right)^\top \\ &= \min_w \sum_{i \in \mathcal{I}_c} w^\top (X_{\text{bold},i} - X_{\text{bold},i+1}) (X_{\text{bold},i} - X_{\text{bold},i+1})^\top w. \end{aligned} \quad (5)$$

Here, $\mathcal{I}_c$ denotes the set of valid consecutive-trial pairs after excluding run and block boundaries. This gives the BOLD variability loss:

$$\mathcal{L}_{\text{bold}}(w) = w^\top C_{\text{bold}} w, \quad C_{\text{bold}} = \sum_{i \in \mathcal{I}_c} (X_{\text{bold},i} - X_{\text{bold},i+1}) (X_{\text{bold},i} - X_{\text{bold},i+1})^\top. \quad (6)$$

Therefore, C_{bold} penalizes large changes in BOLD activity between consecutive trials.

281 The same low-variability condition was also applied to the beta estimates:

$$\begin{aligned}
 \min_w \sum_{i \in \mathcal{I}_c} \|y_{\beta,i} - y_{\beta,i+1}\|_2^2 &= \min_w \sum_{i \in \mathcal{I}_c} (y_{\beta,i} - y_{\beta,i+1}) (y_{\beta,i} - y_{\beta,i+1})^\top \\
 &= \min_w \sum_{i \in \mathcal{I}_c} (w^\top X_{\beta,i} - w^\top X_{\beta,i+1}) (w^\top X_{\beta,i} - w^\top X_{\beta,i+1})^\top \\
 &= \min_w \sum_{i \in \mathcal{I}_c} w^\top (X_{\beta,i} - X_{\beta,i+1}) (X_{\beta,i} - X_{\beta,i+1})^\top w. \quad (7)
 \end{aligned}$$

282 This gives the beta variability loss:

$$\mathcal{L}_\beta(w) = w^\top C_\beta w, \quad C_\beta = \sum_{i \in \mathcal{I}_c} (X_{\beta,i} - X_{\beta,i+1}) (X_{\beta,i} - X_{\beta,i+1})^\top. \quad (8)$$

283 Therefore, C_β penalizes large changes in beta values between consecutive trials.

284 Both BOLD and beta variability terms were included because they capture complementary sources
 285 of variability. The beta estimates reflect activation related to movement execution, so $\mathcal{L}_\beta(w)$ constrains
 286 variability in the movement-related neural response. However, the task also includes other processes.
 287 These sources of variability can be encoded in the BOLD signal even when they are not fully captured by
 288 movement-related beta estimates. Therefore, $\mathcal{L}_{\text{bold}}(w)$ was also included to account for broader task-related
 289 variability encoded in the brain.

290 **General loss function:** After defining the four conditions, the penalty terms were combined into a single
 291 total loss. The total penalty was defined as

$$\begin{aligned}
 \mathcal{L}_{\text{total}}(w) &= \alpha_1 \mathcal{L}_{\text{task}}(w) + \alpha_2 \mathcal{L}_{\text{bold}}(w) + \alpha_3 \mathcal{L}_\beta(w) + \alpha_4 \mathcal{L}_{\text{smooth}}(w) \\
 &= w^\top C_{\text{total}} w, \quad (9)
 \end{aligned}$$

292 where

$$C_{\text{total}} = \alpha_1 C_{\text{task}} + \alpha_2 C_{\text{bold}} + \alpha_3 C_\beta + \alpha_4 C_{\text{smooth}}. \quad (10)$$

293 The parameters α_1 , α_2 , α_3 , and α_4 control the relative contributions of task engagement, BOLD variability,
 294 beta variability, and spatial smoothness, respectively.

295 The final objective combined this total penalty with behavioural relevance. The model minimized the
 296 regularization penalties while maximizing the association between the projected beta signal and reaction
 297 time:

$$\min_w \frac{\gamma w^\top C_{\text{total}} w - w^\top C_b w}{w^\top C_d w} \quad \text{s.t.} \quad \|w\|_2 = 1. \quad (11)$$

298 Here, γ controls the balance between satisfying the regularization criteria and maximizing behavioural
 299 relevance. The denominator $w^\top C_d w$ normalizes the objective by the variance of the projected beta signal.

300 The resulting projection was defined as

$$y = w^\top X_\beta. \quad (12)$$

301 The optimization problem was solved under the unit-norm constraint on w using the Sequential Least
 302 Squares Programming (SLSQP) algorithm (Nocedal and Wright, 2006; Kraft, 1988). Trials were divided
 303 into contiguous folds for cross-validation. For each fold, the optimization was solved independently, and
 304 the resulting weights were projected back from PCA space to voxel space. Final voxel maps and projected
 305 signals were obtained by averaging across folds. The resulting weight assigned to each voxel reflected
 306 its contribution to satisfying the combined criteria of task engagement, behavioural relevance, spatial

smoothness, and low trial-to-trial variability. Voxels with the largest weights were then defined as the target network.

Finally, we defined the vigour network as voxels whose weights exceeded the 90th percentile. As medication state and GVS condition were excluded from the optimization procedure, we then assessed whether either variable modulated this network.

2.7 Medication-state network analysis

We tested whether the vigour network showed similar organization across subjects in the OFF- and ON-medication states. For each subject and session, we constructed an ROI-level network using the voxels retained by the vigour network. Voxels were assigned to anatomical ROI families using the Harvard atlas. The network comprised 15 ROI families: Amygdala, Cerebellum, Cingulate Cortex, Inferior Frontal Gyrus, Dorsolateral Prefrontal Cortex, vmPFC/dmPFC, Parietal Cortex, Precentral Cortex, Temporal Cortex, Thalamus, Occipital Cortex, Insular/Opercular Cortex, Hippocampus, Basal Ganglia, and other Cerebral Cortex.

Each ROI family was divided into left- and right-hemisphere nodes according to MNI coordinates. Voxels with $|x| \leq 1$ mm were treated as midline voxels and assigned to the hemisphere containing the larger number of selected voxels for that ROI. When the two hemispheres contained equal numbers, the voxel was assigned to the left hemisphere. Hemisphere-specific ROI nodes containing fewer than five selected voxels were omitted. In the final optimized voxel set, this procedure yielded the same full set of hemisphere-specific nodes for all subjects and sessions, corresponding to 496 unique undirected edges.

For each subject and session, we computed a trial-wise ROI beta series by averaging voxel-wise beta values within each node. Pairwise ROI connectivity was quantified using mutual information (MI), estimated with the Kraskov-Stoegbauer-Grassberger (KSG) estimator (Kraskov et al., 2004), with $k = 3$ and numerical jitter of 1×10^{-10} . We selected the KSG estimator because the trial-wise beta series is a continuous-valued signal, and KSG estimates MI directly in the continuous domain via k -nearest-neighbor distances in the joint and marginal spaces. This avoids the discretization step required by histogram- or binning-based MI estimators, which discards within-bin variability, introduces sensitivity to bin width and placement, and biases estimates in the limited sample of trial-wise fMRI data. As a nonparametric estimator, KSG additionally captures both linear and nonlinear statistical dependencies without assuming a specific functional form (e.g., Gaussianity) for the joint distribution. Intuitively, MI quantifies how much knowing the response amplitude in one ROI reduces uncertainty about the response in another. Because MI is a symmetric, undirected measure of statistical dependence, the resulting connectivity values should not be interpreted as reflecting excitation, inhibition, or directed influence.

For each session-specific connectivity matrix, the diagonal was set to zero, and all off-diagonal edge weights were retained without thresholding. KSG mutual-information estimates were stored in nats and were not min-max normalized or otherwise rescaled before Laplacian construction. Each weighted connectivity matrix C was converted into a normalized graph Laplacian using degree normalization based on absolute edge weights:

$$D_{ii} = \sum_j |C_{ij}|, \quad L = I - D^{-1/2} C D^{-1/2}.$$

For each network, all eigenvalues of L were computed and sorted in ascending order, yielding a 32-value spectral representation of the network. Network dissimilarity between two subject was then quantified as the Euclidean distance between their sorted Laplacian eigenvalue vectors.

Medication-state effects on network organization were tested using between-subject same-state network pairs. Pair classes were defined as OFF-OFF pairs and ON-ON pairs, and we also examined the pooled on and off states. The dependent variable was the Laplacian spectral distance between the two networks in each pair. To account for repeated involvement of the same subjects across multiple pairwise distances, we used a mixed-effects model with pair class as a fixed effect and crossed random intercepts for the two subjects contributing to each pair.

2.8 GVS-related network analysis

To test whether GVS modulated activity within the optimized motor-vigour network, we used the final optimized voxel set from the main analysis. For each subject, medication session, GVS condition, and trial, we constructed condition-specific ROI beta matrices. Rows corresponded to ROI nodes, columns corresponded to trials, and each entry was the mean GLMsingle beta value across selected final-network voxels assigned to that ROI for that trial.

ROI-level GVS effects were quantified as sham-referenced changes in beta amplitude separately for the OFF- and ON-medication sessions. For subject s , medication state $m \in \{\text{OFF}, \text{ON}\}$, ROI r , and active GVS condition c , the change score was defined as

$$\Delta\beta_{s,m,r,c} = \bar{\beta}_{s,m,r,c} - \bar{\beta}_{s,m,r,\text{sham}},$$

where $\bar{\beta}$ denotes the mean across trials. Thus, the OFF table compares each OFF-session GVS condition with OFF-session sham, whereas the ON table compares each ON-session GVS condition with ON-session sham.

For statistical inference, trial-level ROI beta values from each active GVS condition were compared with trial-level ROI beta values from the corresponding same-session sham condition using two-sided Welch t -tests. Welch tests were used to allow unequal variances and unequal trial counts across conditions. False discovery rate correction was applied within each subject and session-specific contrast across the retained ROI-by-condition test family; the tables list ROIs surviving FDR correction at $q < 0.05$ for each active GVS condition.

3 RESULTS

Results are presented in the same order as the study aims. We first tested whether the optimization recovered a behaviour-linked signal with low trial-to-trial variability, then characterized the anatomical distribution of the resulting network, compared it with the task-engagement reference map, and finally examined medication- and GVS-related network effects.

3.1 Optimization identifies a behaviour-linked low-variability signal

The optimization framework was designed to favour projections that were task-engaged, related to reaction time, spatially smooth, and low in trial-to-trial variability. Across ablation settings, the full model (task engagement + BOLD variability + beta variability + spatial smoothness) yielded the lowest mean evaluation score ($S = 0.0164$; Supplementary Figure 1), indicating the best overall trade-off between held-out behavioural association and cross-fold stability in this dataset. The task-only model achieved the highest mean test correlation with reaction time (0.1918) but showed a larger relative train-test loss gap than the full model (0.2844 vs. 0.1257), consistent with poorer generalization. The full model outperformed 8 of the 11 alternative settings after Bonferroni correction, although its advantage over the strongest single-term baselines was limited.

We next asked whether the selected solution showed the expected low-variability profile in analyses that were not themselves part of the model-selection score. At the run level, the optimized projection had lower consecutive-trial variability than reaction time across 56 runs from 14 subjects ($\beta = -43.94$, $z = -16.52$, $p = 2.55 \times 10^{-61}$; Supplementary Figure 2). At the voxel level, selected voxels were overrepresented among low-variability voxels relative to 43,711 motor-area voxels not selected by the optimization across pooled percentile thresholds, and the observed mean variability of the selected set (2.11) was lower than the mean obtained from size-matched resamples of the motor-area non-selected pool (2.95, 95% CI [2.37, 4.44], empirical $p \approx 0.001$; Supplementary Figure 3).

Together, these results support choosing the full model as the most balanced setting among those tested and show that the resulting projection is relatively low-variability compared with behaviour and with a motor-area comparison pool in this dataset. We next examined the spatial distribution of the optimized network.

3.2 Spatial distribution of the optimized vigour-related network

The optimized vigour-related network comprised 2,674 voxels and showed a clear left-hemisphere predominance on the sagittal, coronal, and axial slice overlays in Figure 2. A motor-related core was present in peri-central cortex, superior parietal cortex, the left putamen, and right cerebellar regions. Additional clusters extended into the frontal pole, the middle and inferior temporal cortex, parahippocampal cortex, hippocampus, and amygdala. When normalized by ROI size, coverage was strongest in amygdalar, posterior temporal, hippocampal/parahippocampal, and putaminal regions, while motor-related regions remained prominently represented (Supplementary Table 1).

The optimized map was not a focal primary motor cluster. Instead, it was a distributed network with a motor-related core, but with strongest ROI-normalized representation in temporal-limbic regions and the left putamen. Whether this pattern reflects vigour-specific coding rather than broader task-related activation is tested in the next section.

3.3 Limited overlap with the task-engagement reference map

To compare the optimized vigour-related network with the task-engagement reference map on matched terms, both maps were thresholded using the same percentile-based rule to define the final suprathreshold networks. Under this common threshold, spatial overlap was limited, with 102 shared voxels, a Dice coefficient of 0.049, and a center-of-mass distance of 47.3 mm (Fig. 3; Supplementary Tables 2 and 3). The task-engagement reference map was concentrated primarily in postcentral, occipital, posterior cingulate, central opercular, and thalamic regions, whereas the optimized network showed greater representation in frontal pole, middle and inferior temporal gyri, parahippocampal gyrus, hippocampus, amygdala, superior parietal lobule, and left putamen. Although both maps included voxels within some of the same atlas-defined regions, voxelwise overlap within those regions remained sparse, suggesting that the two maps emphasize different spatial subcomponents.

3.4 Medication-related differences in network topology

Pairwise spectral distances between subject-session connectivity matrices were lowest for ON–ON comparisons, highest for OFF–OFF comparisons, and intermediate for OFF–ON comparisons (Fig. 4). All three pairwise contrasts were significant (OFF–OFF vs ON–ON, $p = 5.5 \times 10^{-6}$; OFF–OFF vs OFF–ON, $p = 0.0034$; ON–ON vs OFF–ON, $p = 0.02$; mixed-effects model). Thus, the selected ROI-edge network was more similar across subjects in the ON state than in the OFF state. The most conservative interpretation is increased cross-subject convergence, or reduced heterogeneity, in the ON state. The ROI-to-ROI edges showing the largest between-session changes across paired subjects are illustrated in Supplementary Figure 4.

We next tested whether the lower ON–ON distances could be explained by a simple ON-state stability bias in the selected voxels. The selected voxel set remained less variable than the motor control pool in both OFF (2.00 vs. 2.52) and ON (2.31 vs. 3.52) sessions, but it did not show preferential stabilization in the ON state. The OFF-to-ON increase in normalized consecutive-trial beta differences was small and nonsignificant for selected voxels ($\Delta_{sel} = +0.308$, paired $t = 0.669$, $p = 0.515$), larger for control voxels ($\Delta_{ctl} = +0.998$, paired $t = 1.813$, $p = 0.093$), and the difference in change between the two sets was significant ($\Delta_{sel} - \Delta_{ctl} = -0.690$, $p = 0.046$; Supplementary Figure 5). Thus, the higher ON–ON similarity was not due to greater stability of the selected voxels in the ON state.

3.5 GVS-related effects in the final network

To assess the effect of GVS on the final network, we compared the mean beta value of each ROI between each active GVS condition and sham, separately within medication state and within subject (Supplementary Tables 4 and 5). Because GVS effects varied across participants (Lee et al., 2021; Menon et al., 2025), these comparisons were evaluated at the individual-subject level rather than interpreted primarily as a uniform group effect. This sham-referenced ROI analysis showed regionally selective changes in mean beta values within the final network, but the pattern was heterogeneous across subjects and stimulation conditions rather than spatially consistent across ROIs. In the OFF medication state, 52 subject-condition-ROI comparisons survived within-subject FDR correction, most often involving precentral, dorsolateral prefrontal, hippocampal, basal ganglia, vmPFC/dmPFC, and occipital ROIs. In the ON medication state,

72 comparisons survived, most commonly in other cerebral cortex, hippocampal, basal ganglia, parietal, dorsolateral prefrontal, and amygdalar ROIs. Many of the ON-state effects were concentrated in a subset of participants, particularly sub-pd012, and were often decreases relative to sham. Overall, these findings indicate that GVS modulated beta responses in parts of the final network, but the effects were regionally selective and strongly subject dependent rather than consistent at the group level.

4 DISCUSSION

This study asked whether trial-wise variability in motor vigour in Parkinson's disease reflects instability in a central vigour-related neural signal, or whether some of the behavioural variability arises downstream of a relatively stable central signal. The results support the second interpretation. Using fMRI during a ballistic hand-squeeze task, we identified a distributed neural projection that tracked reaction time while showing lower consecutive-trial variability than the behavioural signal itself. The selected voxels were also less variable than a broader motor-area comparison pool. Thus, the main finding is not simply that reaction time was associated with brain activity. Rather, the study recovered a behaviour-linked neural expression of vigour in Parkinson's disease that was more stable than the behaviour it tracked.

This dissociation is important because behavioural variability alone cannot reveal where instability enters the motor system. In Parkinson's disease, response times fluctuate from moment to moment, and motor-unit firing can be more variable even when average performance is partly preserved (MacDonald et al., 2025; Wilson et al., 2020). More generally, movement variability can arise from multiple sources, including central preparatory states, adaptive exploration, and downstream motor noise (Churchland et al., 2006; Faisal et al., 2008; Harris and Wolpert, 1998; Pekny et al., 2015; Wu et al., 2014). The present findings therefore argue against a simple interpretation in which variable vigour in Parkinson's disease directly reflects an unstable central vigour command. Instead, they suggest that a relatively stable central signal related to trial-wise vigour can still be recovered in Parkinson's disease, and that at least some behavioural fluctuation in when movements are initiated may arise downstream of this signal, or through its interaction with later motor processes.

A potential concern is that this conclusion could appear circular, because low consecutive-trial variability was one of the optimization criteria. However, the analysis did not merely impose stability and then report the same property back as a result. The model was constrained not only by stability, but also by task engagement, behavioural relevance, and spatial coherence. We therefore tested whether the recovered solution showed a distinctive stability profile relative to meaningful baselines. It did. At the run level, the optimized projection was less variable than reaction time. At the voxel level, the selected voxels were concentrated among low-variability voxels and showed lower mean variability than size-matched samples drawn from non-selected motor-area voxels. These validation analyses show that the solution was not an arbitrary smooth sample of motor activity and was not simply a restatement of behavioural variability. Rather, the projection captured a stable, behaviour-linked neural component whose stability exceeded that of the behavioural signal itself.

The spatial organization of this projection further supports this interpretation. The recovered vigour-related network included a motor-related core, with contributions from pericentral cortex, superior parietal cortex, putamen, and cerebellar regions. This is consistent with previous work implicating motor cortical, basal ganglia, thalamic, and cerebellar systems in movement speed, force, amplitude, and vigour (Anderson et al., 2012; Dudman and Krakauer, 2016; Shirinbayan et al., 2019b; Spraker et al., 2007b; Turner and Desmurget, 2010; Yttri and Dudman, 2016). However, the projection was not limited to execution-related motor regions. It also included the amygdala, hippocampus, parahippocampal cortex, temporal cortex, frontal pole, and putamen. This broader distribution is informative. A signal reflecting pure motor execution would be expected to localize more strongly to primary motor and adjacent execution-related areas. In contrast, the present projection was enriched in temporal-limbic and striatal regions that are well positioned to integrate cue value, context, salience, motivational state, and action readiness.

This network organization fits the behavioural demands of the task. The putamen and broader basal ganglia are directly relevant to vigour because dopaminergic cortico-basal ganglia circuits regulate movement velocity and the energetic investment in action (Dudman and Krakauer, 2016; Niv et al., 2007; Turner and Desmurget, 2010; Yttri and Dudman, 2016; Zénon et al., 2016). The amygdala is also relevant in this context. Primate amygdala neurons encode positive and negative visual-stimulus value during learning (Paton et al.,

2006), and anatomical evidence supports a direct amygdala–motor pathway through which affective or salient cues can shape action (Grèzes et al., 2014). Hippocampal and parahippocampal involvement is similarly meaningful. Reward behaviour depends on hippocampus–nucleus accumbens synaptic strength (LeGates et al., 2018), and the hippocampus can organize upcoming skilled action sequences (Yewbrey and Kornysheva, 2024). Together with evidence that the hippocampus supports visually cued contextual response selection (Kim et al., 2012), these findings suggest that the recovered vigour network integrates motor preparation with the contextual and motivational signals that determine whether an action is initiated quickly on a given trial.

The limited overlap between this projection and the task-engagement reference map strengthens the same argument. A task-engagement map asks where activity is reliably evoked by performing the task. The present optimization asked a different question: which spatially coherent neural projection is linked to how quickly an action is initiated while maintaining a stable expression over consecutive trials? The low overlap indicates that the network most consistently activated by the task is not the same as the network that best captures stable trial-wise vigour-related coding. This distinction is important because it separates mean task activation from trial-wise vigour coding. Vigour is therefore not simply a property of task-evoked activation amplitude. It is expressed through the organization of a distributed neural signal that links preparatory, motivational, contextual, and motor systems.

The medication findings extend this central-versus-downstream interpretation. In the ON medication state, the selected ROI-edge network was more similar across subjects than in the OFF state, with OFF–ON comparisons falling between ON–ON and OFF–OFF comparisons. This pattern indicates that dopaminergic medication increased cross-subject convergence in the organization of the recovered vigour-related network. Crucially, this effect was not explained by a simple increase in local voxel stability. The selected voxel set remained less variable than the motor comparison pool in both medication states, but it did not show preferential stabilization in the ON state. Thus, dopamine did not merely reduce trial-to-trial variability within the selected voxels. It changed the organization of the network.

This provides a stronger interpretation than saying that medication reduced neural noise. The data suggest that dopamine modulates the topology of the central vigour-related signal: the pattern of statistical dependence among network nodes became more consistent across patients when they were ON medication. This is consistent with prior work linking dopamine to movement vigour and with evidence that levodopa can alter functional network organization in Parkinson's disease (Niv et al., 2007; Berman et al., 2016; Zénon et al., 2016; Mazzoni et al., 2007b). It also yields a clear testable prediction. Dopaminergic medication should increase the reproducibility or convergence of vigour-related network topology even when voxel-level stability is controlled, and individual changes in this topology should predict behavioural changes in vigour or response-time variability.

The medication result also shows why a purely downstream-noise account is incomplete. If behavioural variability arose only from noise added after a stable central command, dopaminergic medication might improve behavioural output without systematically reorganizing the central network that tracks trial-wise vigour. Instead, the ON-state result suggests that dopamine acts on the organization of the behaviour-linked central signal itself. Thus, some behavioural variability may arise downstream of central vigour-related coding, but dopamine also appears to shape the central network through which that coding is expressed.

The GVS findings add a different piece to the same story. Active stimulation produced sham-referenced beta changes within parts of the final network, but the pattern was heterogeneous across subjects, stimulation waveforms, medication states, and ROIs. In the OFF state, significant effects most often involved precentral, dorsolateral prefrontal, hippocampal, basal ganglia, vmPFC/dmPFC, and occipital ROIs. In the ON state, effects more often involved other cerebral cortex, hippocampal, basal ganglia, parietal, dorsolateral prefrontal, and amygdalar ROIs. Many ON-state effects were concentrated in a subset of participants and often reflected decreases relative to sham. The value of this result is not that GVS restored a common vigour network configuration. It did not. Instead, GVS showed that components of the recovered vigour-related network are perturbable by vestibular stimulation, but that this perturbation is state-dependent and subject-specific. This interpretation is consistent with previous work showing that GVS effects in Parkinson's disease depend on stimulation frequency, individual physiology, and parameter selection (Lee et al., 2021; Cai et al., 2018). The present data therefore separate two ideas that should not be conflated: a network can be modulated by stimulation without being reorganized into a common, dopaminergically supported configuration.

This distinction has practical implications for future stimulation work. The present findings suggest that GVS studies should move beyond testing whether stimulation changes mean activation in selected ROIs. A more informative approach would ask whether individualized stimulation parameters can alter the stability, topology, or behavioural coupling of the vigour-related network. In this framework, the relevant target is not simply a region that responds to stimulation, but a network property that links stimulation-induced neural change to trial-wise vigour.

Several limitations define the scope of this interpretation. First, the study did not include a healthy control group. The results therefore cannot determine whether the recovered network is specific to Parkinson's disease or whether it reflects a more general vigour-related signal that remains detectable in Parkinson's disease. A control group would also help determine whether the neural-behavioural stability dissociation is exaggerated, preserved, or qualitatively altered in Parkinson's disease. Second, reaction time was used as the primary trial-wise index of vigour because it was the most robust behavioural measure in this dataset. Reaction time is highly relevant to response initiation, but vigour also includes movement speed, force amplitude, force rise time, and movement duration. Future studies should test whether the same network tracks multiple components of vigour, or whether distinct but overlapping networks support initiation speed and force production.

A further limitation is that the optimization framework recovered the network through a linear projection of ROI activity. This approach is interpretable and well suited to identifying a consistent low-variability axis across trials, but the relationship between distributed brain activity and trial-wise vigour is unlikely to be strictly linear. Neural contributions to movement initiation may combine multiplicatively, saturate at high activation levels, or depend on interactions between regions that a weighted sum cannot capture. The linear constraint therefore limits the model to the component of the brain-behaviour relationship that can be expressed as a fixed linear combination of ROI signals. Any nonlinear structure is necessarily absorbed into the residual rather than reflected directly in the recovered network. Future work should test whether nonlinear approaches, such as kernel-based projections, manifold-learning methods, or neural-network models trained against the same behavioural target, recover a similar set of regions or reveal additional structure that the linear formulation cannot express. Convergence across linear and nonlinear models would strengthen the claim that the network reflects a genuine vigour-related substrate rather than a feature of the modelling choice.

The ROI-level connectivity analyses also relied on mutual information, which is a symmetric and undirected measure of statistical dependence. Although this estimator captures both linear and nonlinear relationships without distributional assumptions, it cannot distinguish driving from driven regions or characterize the direction of information flow within the network. The medication-related differences in network topology should therefore be interpreted as changes in the overall pattern of statistical dependence, not as changes in directed influence between specific nodes. Future work should apply directional approaches, such as Granger causality, dynamic causal modelling, or transfer entropy, to test whether particular nodes preferentially drive trial-wise vigour fluctuations. Such analyses could also test whether directional or graph-theoretic features of the network predict clinical severity, including MDS-UPDRS motor scores or bradykinesia subscores.

Finally, the same dataset was used both to identify the vigour-related network and to characterize its properties. The optimization used cross-validated folds, and several supporting analyses, including the comparison of consecutive-trial variability against behaviour, the comparison with a motor-area pool, the medication-state network similarity analysis, and the GVS analyses, relied on metrics that were not part of the model-selection score. Nevertheless, the absence of external validation in an independent sample limits inferences about generalizability and leaves open the possibility that some features of the recovered network reflect sample-specific structure. Replication in an independent Parkinson's disease cohort, in healthy adults performing the same task, and in Parkinson's disease patients performing a different motor task will be needed to determine whether the same network emerges across samples, tasks, and analytic pipelines. The GVS analyses should also be considered exploratory and subject-level by design. They show that the network is sensitive to vestibular stimulation, but they do not identify a uniform stimulation target or therapeutic waveform. Future work should therefore test individualized stimulation parameters and evaluate whether GVS changes the same network-level properties implicated here: trial-wise stability, behavioural coupling, and cross-subject or within-subject network organization.

Overall, the present study suggests that trial-wise vigour variability in Parkinson's disease should not be interpreted simply as instability of a central motor command. A relatively stable, behaviour-linked neural signal can be recovered, and this signal is organized across motor, striatal, limbic, contextual, and motivational systems. Behavioural variability may therefore partly arise downstream of central vigour-related coding, but dopamine also appears to shape the network through which this coding is expressed. This places vigour variability in Parkinson's disease between two mechanisms: a recoverable central signal that remains more stable than behaviour, and downstream or state-dependent processes that determine how consistently that signal is translated into action.

5 CONCLUSIONS

The present study identifies a distributed vigour-related neural projection in Parkinson's disease whose trial-wise expression is more stable than the behavioural reaction-time fluctuations it tracks. The anatomy of this projection links motor and striatal systems with temporal–limbic regions involved in value, context, salience, and action organization. Dopaminergic medication increased cross-subject convergence of this network without simply stabilizing the selected voxels, suggesting that dopamine acts on the organization of central vigour-related coding. Together, these findings support a model in which PD-related variability in response initiation is not solely the product of an unstable central signal. Instead, a relatively stable central vigour-related signal can be recovered, while part of the behavioural variability may arise downstream as that signal is transformed into motor output.

ETHICS STATEMENT

The studies involving human participants were reviewed and approved by the Clinical Research Ethics Board at the University of British Columbia. The participants provided written informed consent to participate in this study.

CONFLICT OF INTEREST STATEMENT

[Please confirm: The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.]

AUTHOR CONTRIBUTIONS

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SUPPLEMENTAL DATA

Supplementary material for this article is provided in the accompanying Supplementary Material file.

DATA AVAILABILITY STATEMENT

[Please complete: The datasets generated and/or analyzed during the current study are available in repository name/link, or state the appropriate restriction if the data cannot be publicly shared.]

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FIGURE CAPTIONS

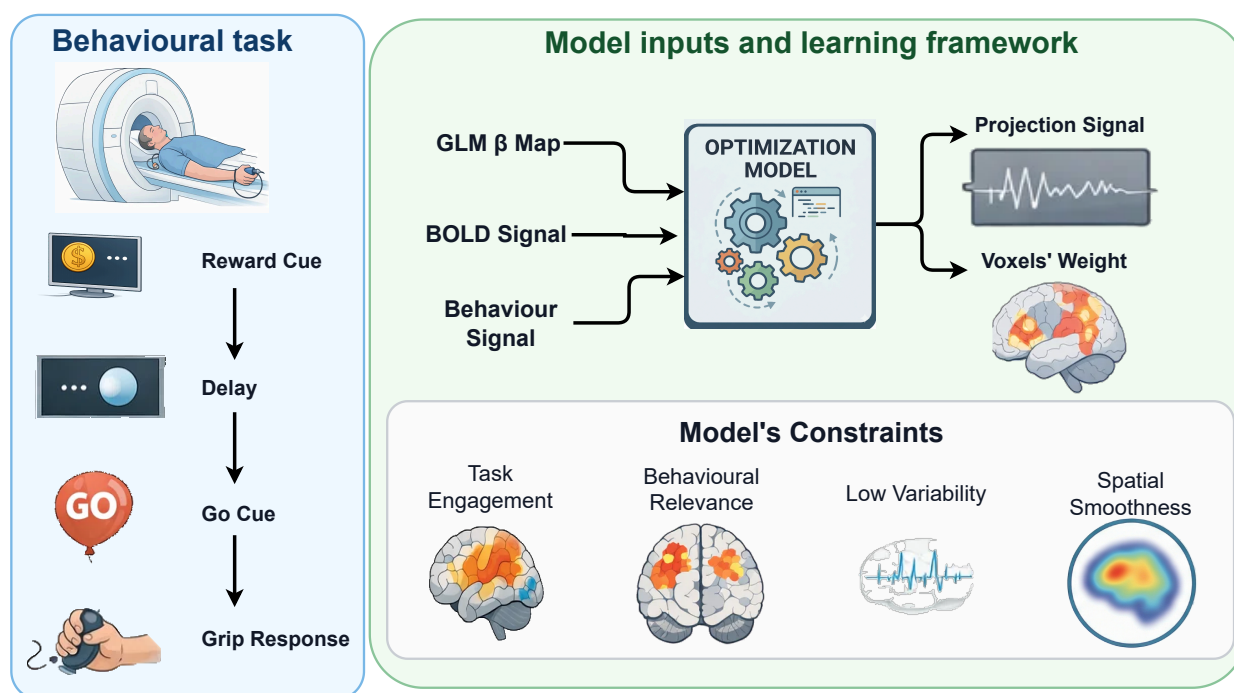


Figure 1. Overview of the task and analysis pipeline. Participants perform a ballistic hand-squeeze task during fMRI, with each trial progressing from a reward cue to a delay, then a go cue, followed by the grip response. During this stage, both behavioural data and BOLD fMRI data are collected simultaneously. Trial-wise GLM beta maps, BOLD signals, and behavioural reaction-time measures were entered into a constrained optimization framework designed. The optimization searches for a neural projection that is task-related, behaviourally relevant, spatially smooth, and relatively stable across consecutive trials. The outputs are a projection signal, representing the low-dimensional trial-wise network response, and a voxel-weight map, indicating which brain voxels contribute most strongly to the final vigour-related network.

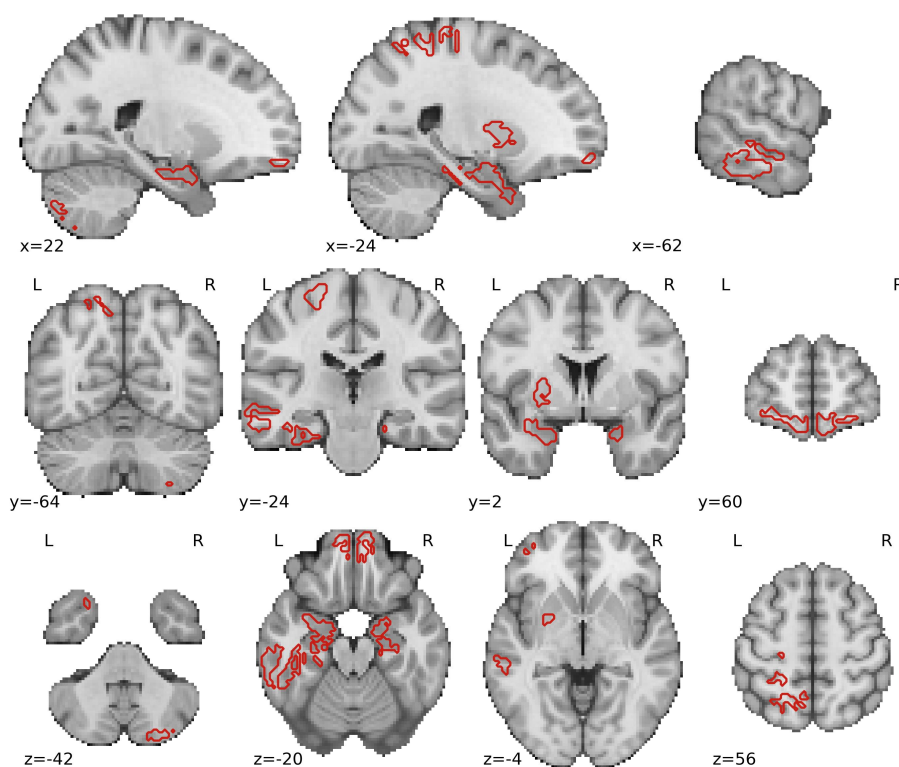


Figure 2. Spatial distribution of the identified candidate network shown on sagittal, coronal, and axial slices in MNI space. Red contours highlight suprathreshold voxels, overlaid on the MNI152 anatomical template. The selected slices capture all connected components of the network.

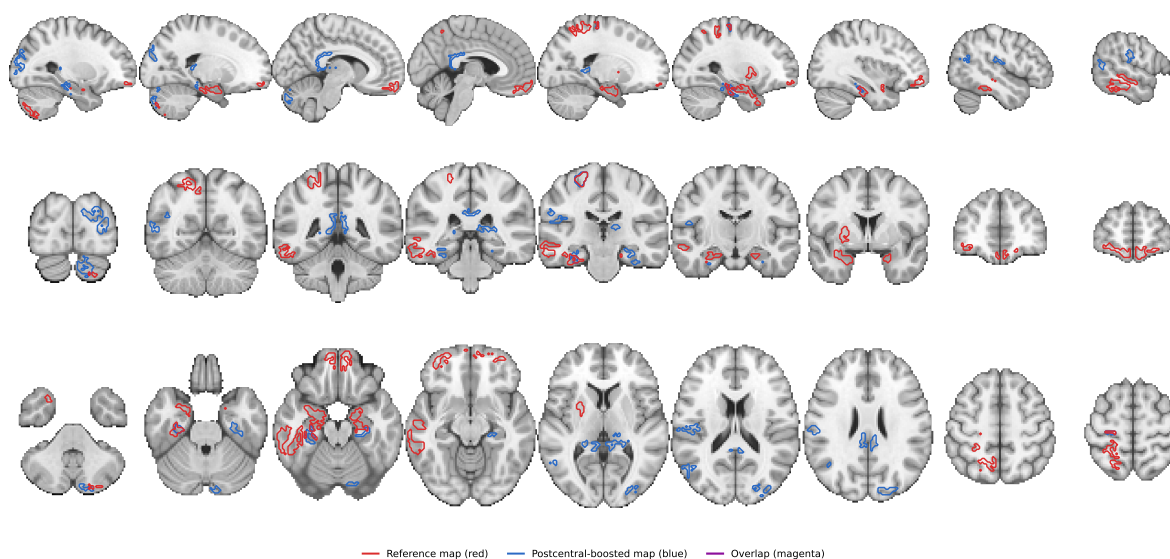


Figure 3. Comparison of the all-constraints vigour-related network and the task-engagement reference map in MNI space. Red contours show the network obtained from the full optimization objective, blue contours show the reference network, and magenta contours indicate overlapping voxels.

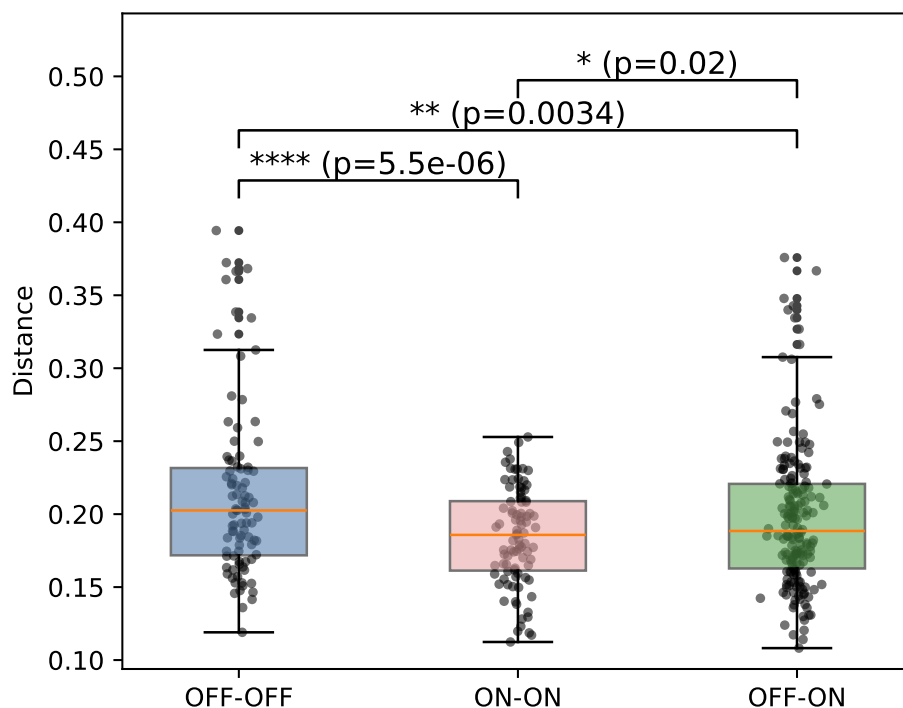


Figure 4. Cross-subject similarity of the selected ROI-edge network estimated with KSG mutual information. Pairwise distances were computed between subject-session connectivity matrices using the signed normalized Laplacian spectral distance. Only cross-subject comparisons are shown ($n = 14$ subjects: OFF-OFF $n = 91$, ON-ON $n = 91$, OFF-ON $n = 182$). Each dot represents one pairwise comparison, and boxplots show the median and interquartile range. ON-ON pairs showed the lowest distances, OFF-OFF pairs the highest, and OFF-ON pairs were intermediate (mixed-effects model: OFF-OFF vs ON-ON, $p = 5.5 \times 10^{-6}$; OFF-OFF vs OFF-ON, $p = 0.0034$; ON-ON vs OFF-ON, $p = 0.02$).